# Term 1 — *"My Math-Fact Smoke Detector"*

### 8 session scripts for Coach Dad to run with Emeka

🔭 **Mode:** Foresight (*catch the problem before it breaks*) · **Pillar:** Yourself **Academic targets baked in:** +/− facts within 20 to automatic recall · explaining her thinking · writing **3-4 neat sentences** (capitals + punctuation). **The real-world problem she's getting ahead of:** *forgetting a math fact until a test catches her.*

> **What she builds:** her own little "Smoke Detector" — a tool that spots the math facts she's *about to forget* (the "smoke") so she can practice them *before* they cause a bad grade (the "fire").

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## How to run these (read once)

- **Length:** each session works in **~30-45 min**. Short is fine — consistency beats length.
- **The fixed rhythm (every session):**

1. **Circle + Callback** (2 min) — one-word check-in, then *retrieve last time's thinking*.
2. **Math reps** (5 min) — the day's fact practice (cards or the app — *emeka-grades* Fast Facts / Math Sprint).
3. **The Move** (5 min) — today's named detective move.
4. **Build / Think** (15-20 min) — she directs, you help.
5. **Share + Log + Open Loop** (5 min) — she shows it, writes one neat log line, takes a question out the door.

- **The two rules that never bend:**
- **Open with a Callback** (make her remember last time). **Close with an Open Loop** (a real question she carries out the door).
- **She decides. You help.** Protect the seed — the ideas are hers.
- **Kit:** a notebook (**"Emeka's Expedition Log"**), pencil/markers, a deck of +/− flash cards, and

one device with her app. *(Most of this is off-screen — the screen only really comes on in Session 6.)*

- **Tone:** you're not her teacher here, you're her **detective partner.** Wobbly facts aren't bad —

they're *clues*. We're hunting them on purpose.

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## SESSION 1 — MAP IT

**Loop step:** *How does my math-fact brain normally work?* · **The Move:** "Map It" — *cue: "Draw how it works."* 🎯 **Objective:** she looks at her own math-fact memory like a machine — no fixing yet, just seeing.

**Math reps (5 min):** a relaxed round of 10 +/− facts within 20 (cards or app). *No pressure — say out loud: "We're not getting them all right today, we're just feeling which ones are fast."*

**Build / Think:** in her log she **draws her "math brain machine"** — how a fact gets from *learned* to *remembered fast*.

💬 **Say this (slow, curious):**

> "Today we're detectives about your own brain — we're not fixing anything, just looking. Deal?" "When I say a math fact like **7 + 6**, what happens in your head? Walk me through it, super slow." "Some facts pop out *fast*, like you don't even think — which ones are like that for you?" "Some facts make you stop and count — show me one of those. That's not bad! That's a clue." "So your math-fact brain is kind of like a machine. Draw me how it works — where do the fast facts live, where do the slow ones get stuck?"

🗣️ **Share:** she shows her drawing and names **2 fast facts** and **1 slow fact.**

🚪 **Open loop (out the door):**

> "This week, catch yourself **one time** when a math fact makes you stop and think. Remember *exactly* which fact it was — bring it to me."

✍️ **Log line:** *"My math brain is fast at ___ and slow at ___."*

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## SESSION 2 — FIND THE WEAK SPOT

**Loop step:** *Which facts could trip me up?* · **The Move:** "Find the Weak Spot" — *cue: "Where does it wobble?"* 🎯 **Objective:** find her shaky fact families — the weak spots — *before* a test finds them.

🔁 **Callback:**

> "Last time you were going to catch a fact that made you stop. Which one was it? Tell me exactly what happened."

**Math reps / Build (this is the same activity today):** the **Fact Sort.**

> "Engineers find the weak spot *before* the bridge falls. We're finding your wobbly facts *before* a test does. I'll say a fact — if it comes out fast, say **'fast.'** If you have to think even a little, say **'wobbly.'** We *want* the wobbly ones — that's the treasure."

Run ~15 facts. Make two piles/lists: **FAST** and **WOBBLY.** Then look together:

> "Look at your wobbly pile. Notice anything? Are they mostly the **+9s**? The **teen take-aways** like 15 − 8? Let's see the pattern."

🚪 **Open loop:**

> "Pick your **#1 wobbliest fact** — that's our target. Say it out loud 3 times before bed each night."

✍️ **Log line:** *"My wobbliest facts are ___. My #1 target is ___."*

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## SESSION 3 — SMOKE BEFORE FIRE

**Loop step:** *What's the early signal?* · **The Move:** "Smoke Before Fire" — *cue: "Smell the smoke."* 🎯 **Objective:** name the "smoke" — the warning sign that a fact is about to be forgotten.

🔁 **Callback:**

> "What was your #1 target fact? Did saying it at night help even a little?"

**Math reps (5 min):** the target fact + 3-4 other wobbly ones.

**The Move + Think:**

> "A smoke detector **beeps before the fire.** So — what's your **'smoke'** for a fact you're about to forget? What does your *body* do right before?" "Do your **fingers come up**? Do you go **quiet**? Do your **eyes look up** at the ceiling? Let's catch your exact signal." "Cool — *that's* your smoke. When you notice THAT, your detector should beep: *'practice this one!'*"

🗣️ **Share:** she acts out her smoke signal.

🚪 **Open loop:**

> "Watch for your smoke signal this week — at school, at homework, anywhere. **Count how many times** you see smoke."

✍️ **Log line:** *"My smoke signal is ___. When I see it, it means ___."*

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## SESSION 4 — PLAY THE TAPE FORWARD

**Loop step:** *If I ignore the smoke, then what?* · **The Move:** "Play the Tape Forward" — *cue: "Then what? Then what?"* 🎯 **Objective:** she *feels* the reason this matters (the "why" that powers the whole project).

🔁 **Callback:**

> "How many times did you see your smoke signal this week? What was happening when you did?"

**Math reps (5 min):** the target facts — notice if any are getting faster.

**The Move:**

> "Let's play the tape forward like a movie. If you keep seeing smoke on **7 + 6** and do *nothing*... then what? ... and then what? ... and then what happens on the test? ... and then how do you feel?" "Now let's **rewind and play the *other* movie.** If you *catch* the smoke and practice 7 + 6 for three days... then what? ... then what happens on the test? ... how do you feel *that* time?" "**Which movie do you want?** ... Good. That's why we're building the detector."

🚪 **Open loop:**

> "Tell **one person at home** which fact you're catching, and *why it matters to you*."

✍️ **Log line (NEAT — this one's writing practice):** *3-4 sentences, capital letters, periods —*

> *"There are two movies. If I ignore the smoke, ___. If I catch the smoke, ___. I want the second movie because ___."*

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## SESSION 5 — SMALLEST EARLY STOP

**Loop step:** *The tiniest thing that prevents it.* · **The Move:** "Smallest That Works" — *cue: "First, not perfect."* 🎯 **Objective:** design the smallest daily action that stops a wobbly fact from becoming a fire.

🔁 **Callback:**

> "Who did you tell? What did they say when you told them why it matters?"

**Math reps (5 min):** target facts — *time them*. Write the time down (we'll beat it later).

**The Move (brainstorm, then shrink):**

> "Let's brainstorm a **TON** of ways to catch smoky facts — fast, silly ones allowed, no judging. Go!" *(write every idea down — aim for 8-10)* "Now circle the **smallest** one you'd *actually* do every single day — small enough you can't say no. Like: *'Every morning I do 5 wobbly facts; any that make smoke go on my RED list.*'"

🚪 **Open loop:**

> "Try your tiny stop **one morning** before next time. Did it fit into your day, or was it too big?"

✍️ **Log line:** *"My smallest daily stop is ___."*

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## SESSION 6 — BUILD THE STOP *(the screen comes on)*

**Loop step:** *Make the detector real.* · **The Move:** "Real Beats Perfect" — *cue: "It only counts when it's real."* 🎯 **Objective:** build the actual **Smoke Detector** — *she directs, you drive.*

🔁 **Callback:**

> "Did your tiny stop fit your morning? What got in the way?"

**Build (15-20 min) — her project, you're the hands.** Make a **Smoke Detector**: a simple page (build it on the device with AI — *she tells you what she wants, you type*) **or** a poster board she designs. It lists her facts; when one makes smoke, she marks it **RED** = "practice me!" Red facts become exactly what she drills in the app's **Fast Facts / Math Sprint.** *Let her pick the colors, the name, the look — ownership is the point.*

💭 **The critique (the heart of it) — when the first version appears:**

> "You're the **boss** of this build — I just type and cut. What should it look like? What color is 'smoke'?" "Do you want to **change** anything here? What's **missing**? Would this actually make **you** want to use it?" *(When she's stuck, don't rescue — ask: "What have you tried? What's one thing you could try next?" then wait.)*

🗣️ **Share:** she shows the family her **live detector** and who it's for (her!).

🚪 **Open loop:**

> "Use your detector **every morning this week.** Bring me the list of which facts turned **red.**"

✍️ **Log line:** *"I built ___. My smoke color is ___."*

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## SESSION 7 — BREAK YOUR OWN FIX

**Loop step:** *Try to make it fail, then patch it.* · **The Move:** "Break Your Own Fix" — *cue: "Be your own villain."* 🎯 **Objective:** make the detector tougher from real use — *and find proof it's working.*

🔁 **Callback:**

> "Which facts turned red this week? … Now the big question: did those red facts get **faster**?"

**Math reps = the PROOF step (5 min):** re-time the facts that were wobbly back in Session 2.

> "Remember our time from Session 5? Let's race it. Faster = proof your detector works."

**The Move (stress-test):**

> "Let's be **sneaky.** Try to make your detector **FAIL.** What day would you forget to use it? When is it annoying? Is there a smoky fact it **misses**?" "What's **one fix** so even sneaky-you can't break it?"

🚪 **Open loop:**

> "**Teach someone else** how your detector works. Bring me what they asked you." *(Teaching it is the deepest test.)*

✍️ **Log line (NEAT — 3-4 sentences, explaining):**

> *"My detector almost broke because ___. I fixed it by ___. I know it works because ___."*

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## SESSION 8 — DEMO DAY

**Loop step:** *Show what you caught before it broke.* · **The Move:** "Catch Yourself Thinking" — *cue: "Name your smartest thought."* 🎯 **Objective:** she presents her detector + the facts she rescued, and names her own smartest thinking.

🔁 **Callback — walk the whole journey together:**

> "Let's tell the whole story: we **Mapped** your brain → found the **weak spots** → named your **smoke** → played the **tape** → picked the **smallest stop** → **built** the detector → and **broke-and-fixed** it. Wow."

**Math reps = the celebration (5 min):** a final timed round of the facts that were wobbly in Session 2.

> "These used to be your wobbly ones. Watch how fast you are now." *(Cheer the before → after.)*

**The Demo (to family):**

> "Show everyone your **Smoke Detector.** Tell them: what problem does it catch? Then show one fact that **used to be wobbly and is now fast.**"

**The Move (metacognition — the real prize):**

> "Last question, and it's the most important one: **the smartest thing I *thought* during this was…?**" *(thought — not did. Help her name a thinking move: "I found the pattern in my wobbly facts," "I caught the smoke before the test," "I made my own fix tougher.")*

🎉 **Close:** celebrate big — a star on the wall, a certificate, her choice. She built a real tool *and* got faster at her facts. **That star is hers.**

🚪 **Open loop → into Term 2 (HOME):**

> "Next time we point this same detective brain at **home** — catching a problem *before* it turns into a fight. Start watching for a moment at home where something's *about to* go wrong."

✍️ **Log line:** *"The smartest thing I thought was ___."*

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## Quick reference — the 8 moves at a glance

| # | Session | The Move | She leaves with… |
|---|---|---|---|
| 1 | Map It | *Draw how it works* | A map of her own math-fact brain |
| 2 | Find the Weak Spot | *Where does it wobble?* | Her list of wobbly facts + #1 target |
| 3 | Smoke Before Fire | *Smell the smoke* | Her personal "about to forget" signal |
| 4 | Play the Tape Forward | *Then what? Then what?* | The *why* (+ a neat-sentence log) |
| 5 | Smallest Early Stop | *First, not perfect* | Her tiny daily catch-it routine |
| 6 | Build the Stop | *Real beats perfect* | A live Smoke Detector she made |
| 7 | Break Your Own Fix | *Be your own villain* | A tougher detector + proof it works |
| 8 | Demo Day | *Name your smartest thought* | Faster facts, a finished build, pride |

*Privacy: first name + age only. Her last name, school, and records stay in Dad's private files.*